# Subspaces and Rank

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## Rank

**Definition 0.1** (Linearly independent vectors)**.** Vectors $\underset{\sim}{v}_1, \ldots, \underset{\sim}{v}_k$ of the same length $p$ are **linearly independent** [1] if the only numbers $c_1, \ldots, c_k$ with

$$c_1 \underset{\sim}{v}_1 + \cdots + c_k \underset{\sim}{v}_k = \underset{\sim}{0}$$

are

$$\begin{aligned} c_1 &= \cdots \\ &= c_k \\ &= 0. \end{aligned}$$

Vectors that are not linearly independent are **linearly dependent**.

**Example 0.1** (Independent and dependent pairs of vectors)**.**

- $\underset{\sim}{v}_1 = (1, 0)$ and $\underset{\sim}{v}_2 = (1, 1)$ are linearly independent: $c_1 \underset{\sim}{v}_1 + c_2 \underset{\sim}{v}_2 = (c_1 + c_2, c_2)$, which is $\underset{\sim}{0}$ only if $c_2 = 0$ and then $c_1 = 0$.
- $\underset{\sim}{v}_1 = (1, 2)$ and $\underset{\sim}{v}_2 = (2, 4)$ are not: $2\underset{\sim}{v}_1 - \underset{\sim}{v}_2 = \underset{\sim}{0}$.

**Exercise 0.1** (Testing whether two features and their total are linearly independent)**.** A table of data that stores two features and also their total holds redundant information, which the test below detects. Let

$$\underset{\sim}{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \qquad \underset{\sim}{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \qquad \underset{\sim}{v}_3 = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}.$$

Are $\underset{\sim}{v}_1$, $\underset{\sim}{v}_2$ and $\underset{\sim}{v}_3$ linearly independent?

**Hint**

Compare $\underset{\sim}{v}_3$ with $\underset{\sim}{v}_1 + \underset{\sim}{v}_2$.

Solution

*Solution* 0.1. No. $\underset{\sim}{v}_1 + \underset{\sim}{v}_2 = (1, 3, 1)^\top = \underset{\sim}{v}_3$, so

$$1 \cdot \underset{\sim}{v}_1 + 1 \cdot \underset{\sim}{v}_2 + (-1) \cdot \underset{\sim}{v}_3 = \underset{\sim}{0},$$

with numbers $c_1 = 1$, $c_2 = 1$ and $c_3 = -1$ that are not all 0. By Definition 0.1, the three vectors are linearly dependent.

**Definition 0.2** (Rank)**.** The **rank** [2] of a matrix $\mathbf{A}$, written rank($\mathbf{A}$), is the largest number of columns of $\mathbf{A}$ that are linearly independent (Definition 0.1).

**Example 0.2** (The rank of two $3 \times 2$ matrices)**.** The matrix $\begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$ has rank 2: if $c_1(1, 1, 1) + c_2(1, 2, 3) = \underset{\sim}{0}$, then subtracting the first entry from the second gives $c_2 = 0$, and then $c_1 = 0$.

The matrix $\begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 1 & 2 \end{bmatrix}$ has rank 1: its second column is twice its first, so the two columns are not linearly independent, but the first column on its own is.

**Definition 0.3** (Full column rank)**.** An $n \times p$ matrix has **full column rank** if its rank (Definition 0.2) is $p$, so that all of its columns are linearly independent.

Remark (Full column rank needs at least as many rows as columns)

*Remark* 0.1 (Full column rank needs at least as many rows as columns). In Example 0.2, the first matrix has full column rank: its rank is 2, and it has 2 columns. The second does not, because its rank is 1.

An $n \times p$ matrix can have full column rank only when $p \leq n$, because more than $n$ vectors of length $n$ are never linearly independent. For example, the $2 \times 3$ matrix $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ has 3 columns of length 2, and $1 \cdot (1, 0) + 1 \cdot (0, 1) - 1 \cdot (1, 1) = \underset{\sim}{0}$, with coefficients that are not all zero. So its 3 columns are not linearly independent, and its rank is less than 3.

**Exercise 0.2** (Finding the rank of a $3 \times 3$ matrix)**.** The rank of a data matrix counts how many of its columns carry new information. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix}.$$

Find rank($\mathbf{A}$).

**Hint**

Add the first two columns of $\mathbf{A}$ and compare them with the third. Then check whether the first two are linearly independent.

Solution

*Solution* 0.2. rank($\mathbf{A}$) = 2. The third column of $\mathbf{A}$ adds up the first two: $(1,0,2)^\top + (0,1,1)^\top = (1,1,3)^\top$, so the three are not linearly independent. The first two are: if $c_1(1,0,2)^\top + c_2(0,1,1)^\top = \underset{\sim}{0}$, then the first entry gives $c_1 = 0$, and the second gives $c_2 = 0$. By Definition 0.2, the rank is 2.

**Exercise 0.3** (Testing whether an all-ones column and one feature have full column rank)**.** Regression software often puts an all-ones column in the input matrix, beside the features. With one feature, measured at 2, 4 and 6, the input matrix is

$$\mathbf{X} = \begin{bmatrix} 1 & 2 \\ 1 & 4 \\ 1 & 6 \end{bmatrix}.$$

Does $\mathbf{X}$ have full column rank?

**Hint**

By Definition 0.3, check whether the two columns of $\mathbf{X}$ are linearly independent.

Solution

*Solution* 0.3. Yes. Suppose $c_1(1,1,1)^\top + c_2(2,4,6)^\top = \underset{\sim}{0}$. The first two entries give

$$c_1 + 2c_2 = 0 \quad \text{and} \quad c_1 + 4c_2 = 0.$$

Subtracting the first equation from the second gives $2c_2 = 0$, so $c_2 = 0$, and then $c_1 = 0$. So the two columns of $\mathbf{X}$ are linearly independent, the rank of $\mathbf{X}$ is 2, and $\mathbf{X}$ has full

column rank.

## Subspaces

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**Definition 0.1** (Subspace, linear subspace)**.** A set $\mathcal{S}$ of vectors in $\mathbb{R}^p$ is a **subspace** [4] of $\mathbb{R}^p$ if it is not empty and it is closed under vector addition and scalar multiplication:

1. if $\underset{\sim}{u} \in \mathcal{S}$ and $\underset{\sim}{v} \in \mathcal{S}$, then $\underset{\sim}{u} + \underset{\sim}{v} \in \mathcal{S}$;
2. if $\underset{\sim}{u} \in \mathcal{S}$ and $c \in \mathbb{R}$, then $c\underset{\sim}{u} \in \mathcal{S}$.

A subspace is also called a **linear subspace**.

**Example 0.1** (A line through the origin is a subspace)**.** Let $\mathcal{S} = \{c\,(1, 2) : c \in \mathbb{R}\}$, the line in $\mathbb{R}^2$ through $(0, 0)$ and $(1, 2)$.

- **Addition:** $a\,(1,2) + b\,(1,2) = (a+b)\,(1,2)$, which is in $\mathcal{S}$.
- **Scalar multiplication:** $c\,(a\,(1,2)) = (ca)\,(1,2)$, which is in $\mathcal{S}$.

So $\mathcal{S}$ is a subspace of $\mathbb{R}^2$. Two other subspaces of $\mathbb{R}^p$ are the set $\{\underset{\sim}{0}\}$ that holds only the zero vector, and $\mathbb{R}^p$ itself.

**Example 0.2** (A line that misses the origin is not a subspace)**.** Let $\mathcal{T} = \{(x, 1) : x \in \mathbb{R}\}$, the horizontal line in $\mathbb{R}^2$ at height 1. The vectors $(0, 1)$ and $(2, 1)$ are in $\mathcal{T}$, but their sum $(2, 2)$ is not, because its second entry is 2, not 1. So $\mathcal{T}$ is not closed under addition, and it is not a subspace.

**Exercise 0.1** (Testing whether vectors with a nonnegative second entry are a subspace)**.** A constraint such as "this entry is at least 0" cuts out a set of vectors, and it is worth knowing whether that set is a subspace. Let

$$\mathcal{H} = \{(x, y) \in \mathbb{R}^2 : y \geq 0\},$$

the vectors whose second entry is nonnegative.
Is $\mathcal{H}$ a subspace of $\mathbb{R}^2$?

**Hint**

Check condition 2 of Definition 0.1 with $c = -1$.

Solution

*Solution* 0.1. No. The vector $(0, 1)^\top$ is in $\mathcal{H}$, but $(-1) \cdot (0, 1)^\top = (0, -1)^\top$ is not, because its second entry is negative. So $\mathcal{H}$ fails condition 2 of Definition 0.1, and it is not a subspace. $\mathcal{H}$ does satisfy condition 1, which shows that one condition alone is not enough.

**Theorem 0.1** (Every subspace contains the zero vector)**.** *If $\mathcal{S}$ is a subspace of $\mathbb{R}^p$ (Definition 0.1), then $\underset{\sim}{0} \in \mathcal{S}$.*

Proof

*Proof.* A subspace is not empty, so it contains some vector $\underset{\sim}{u}$. Closure under scalar multiplication with $c = 0$ puts $0 \cdot \underset{\sim}{u}$ in $\mathcal{S}$, and $0 \cdot \underset{\sim}{u} = \underset{\sim}{0}$. □

**Example 0.3** (Using the zero vector to rule out a subspace)**.** The plane $\{(x, y, z) : x + y + z = 1\}$ in $\mathbb{R}^3$ does not contain $\underset{\sim}{0} = (0, 0, 0)$, because $0 + 0 + 0 = 0 \neq 1$. So, by Theorem 0.1, it is not a subspace. The line $\mathcal{T}$ in Example 0.2 fails the same test: $(0, 0)$ is not in $\mathcal{T}$, because its second entry is not 1.
Containing $\underset{\sim}{0}$ is necessary but not sufficient. The union of the two coordinate axes in $\mathbb{R}^2$, $\{(x, 0) : x \in \mathbb{R}\} \cup \{(0, y) : y \in \mathbb{R}\}$, contains $\underset{\sim}{0}$, but $(1, 0) + (0, 1) = (1, 1)$ lies on neither axis, so the union is not closed under addition and is not a subspace.

**Exercise 0.2** (Testing whether $p_1 + p_2 = 1$ defines a subspace)**.** A coin toss can land heads or tails, and the probabilities $p_1$ and $p_2$ of those two results satisfy $p_1 + p_2 = 1$. That condition defines the set

$$\mathcal{P} = \{(p_1, p_2) \in \mathbb{R}^2 : p_1 + p_2 = 1\}.$$

Is $\mathcal{P}$ a subspace of $\mathbb{R}^2$?

**Hint**

Use Theorem 0.1.

Solution

*Solution* 0.2. No. The zero vector $(0, 0)^\top$ is not in $\mathcal{P}$, because $0 + 0 = 0 \neq 1$. By Theorem 0.1, every subspace contains $\underset{\sim}{0}$, so $\mathcal{P}$ is not a subspace.

**Definition 0.2** (Span)**.** The **span** [5] of vectors $\underset{\sim}{v}_1, \ldots, \underset{\sim}{v}_k \in \mathbb{R}^p$ is the set of all their linear combinations:

$$\operatorname{span}\{\underset{\sim}{v}_1, \ldots, \underset{\sim}{v}_k\} \stackrel{\text{def}}{=} \{c_1\underset{\sim}{v}_1 + \cdots + c_k\underset{\sim}{v}_k : c_1, \ldots, c_k \in \mathbb{R}\}.$$

The vectors $\underset{\sim}{v}_1, \ldots, \underset{\sim}{v}_k$ **span** a set $\mathcal{S}$ if their span equals $\mathcal{S}$.
By convention, the span of the empty list ($k = 0$) is $\{\underset{\sim}{0}\}$, and the empty list counts as linearly independent (Definition 0.1), since it has no coefficients that could fail to be zero.

**Example 0.4** (Two spans in $\mathbb{R}^3$)**.**

- $\operatorname{span}\{(1, 0, 0), (0, 1, 0)\}$ is the set of vectors $(c_1, c_2, 0)$, the plane $z = 0$ in $\mathbb{R}^3$.
- $\operatorname{span}\{(1, 2, 3), (2, 4, 6)\}$ is only the line $\{c\,(1, 2, 3) : c \in \mathbb{R}\}$, because $(2, 4, 6) = 2\,(1, 2, 3)$, so $c_1(1, 2, 3) + c_2(2, 4, 6) = (c_1 + 2c_2)\,(1, 2, 3)$. Adding a vector to a list does not always enlarge its span.

**Exercise 0.3** (Testing whether a vector lies in the span of two others)**.** Whether a target can be written as a linear combination of a few given vectors is a question about a span. Let

$$\underset{\sim}{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \qquad \underset{\sim}{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \qquad \underset{\sim}{w} = \begin{bmatrix} 3 \\ 5 \\ 2 \end{bmatrix}.$$

Is the vector $\underset{\sim}{w}$ in $\operatorname{span}\{\underset{\sim}{v}_1, \underset{\sim}{v}_2\}$?

**Hint**

Look for numbers $c_1$ and $c_2$ with $c_1\underset{\sim}{v}_1 + c_2\underset{\sim}{v}_2 = \underset{\sim}{w}$, one entry at a time.

Solution

*Solution* 0.3. Yes. Entry by entry, $c_1\underset{\sim}{v}_1 + c_2\underset{\sim}{v}_2 = (c_1,\ c_1 + c_2,\ c_2)^\top$. Matching $\underset{\sim}{w}$, the first entry gives $c_1 = 3$ and the third gives $c_2 = 2$. Then the second entry is $3 + 2 = 5$, which matches too, so $\underset{\sim}{w} = 3\underset{\sim}{v}_1 + 2\underset{\sim}{v}_2$ is in the span.

**Theorem 0.2** (A span is a subspace)**.** *For any vectors $\underset{\sim}{v}_1, \dots, \underset{\sim}{v}_k \in \mathbb{R}^p$, $\operatorname{span}\{\underset{\sim}{v}_1, \dots, \underset{\sim}{v}_k\}$ (Definition 0.2) is a subspace of $\mathbb{R}^p$ (Definition 0.1).*

Proof

*Proof.* The span is not empty: it contains $0 \cdot \underset{\sim}{v}_1 + \cdots + 0 \cdot \underset{\sim}{v}_k = \underset{\sim}{0}$.
**Addition.** Take two vectors in the span, $\underset{\sim}{u} = a_1\underset{\sim}{v}_1 + \cdots + a_k\underset{\sim}{v}_k$ and $\underset{\sim}{w} = b_1\underset{\sim}{v}_1 + \cdots + b_k\underset{\sim}{v}_k$. Then

$$\begin{aligned} \underset{\sim}{u} + \underset{\sim}{w} &= (a_1\underset{\sim}{v}_1 + \cdots + a_k\underset{\sim}{v}_k) + (b_1\underset{\sim}{v}_1 + \cdots + b_k\underset{\sim}{v}_k) && \text{(substitute the two combinations)} \\ &= (a_1 + b_1)\,\underset{\sim}{v}_1 + \cdots + (a_k + b_k)\,\underset{\sim}{v}_k && \text{(regroup, and distribute each } \underset{\sim}{v}_i\text{)} \end{aligned}$$

which is a linear combination of $\underset{\sim}{v}_1, \dots, \underset{\sim}{v}_k$, so it is in the span.
**Scalar multiplication.** For $c \in \mathbb{R}$,

$$\begin{aligned} c\,\underset{\sim}{u} &= c\,(a_1\underset{\sim}{v}_1 + \cdots + a_k\underset{\sim}{v}_k) && \text{(substitute the combination)} \\ &= (ca_1)\,\underset{\sim}{v}_1 + \cdots + (ca_k)\,\underset{\sim}{v}_k && \text{(distribute } c \text{ over the sum)} \end{aligned}$$

which is again in the span. □

**Example 0.5** (The line in Example 0.1 is a span)**.** The line $\mathcal{S} = \{c\,(1,2) : c \in \mathbb{R}\}$ in Example 0.1 is span$\{(1,2)\}$, so Theorem 0.2 gives a second proof that it is a subspace. Likewise, the plane $z = 0$ in Example 0.4 is a subspace of $\mathbb{R}^3$, because it is span$\{(1,0,0),(0,1,0)\}$.

**Exercise 0.4** (Recognizing a set of vectors as a span)**.** Describing a set by free parameters often shows that it is a span. Let

$$\mathcal{S} = \{(a+b,\ a-b,\ 2a) : a, b \in \mathbb{R}\}.$$

Is $\mathcal{S}$ a subspace of $\mathbb{R}^3$?

**Hint**

Write $(a+b,\ a-b,\ 2a)$ as $a$ times one vector plus $b$ times another, then use Theorem 0.2.

Solution

*Solution* 0.4. Yes. For every $a, b \in \mathbb{R}$,

$$\begin{bmatrix} a+b \\ a-b \\ 2a \end{bmatrix} = a \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} + b \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix},$$

so $\mathcal{S} = \text{span}\left\{(1,1,2)^\top, (1,-1,0)^\top\right\}$. By Theorem 0.2, $\mathcal{S}$ is a subspace of $\mathbb{R}^3$.

**Theorem 0.3** (A matrix-vector product combines the columns)**.** *If* $\mathbf{A}$ *is an* $m \times n$ *matrix with columns* $\underset{\sim}{a}_1, \ldots, \underset{\sim}{a}_n$ *and* $\underset{\sim}{x} \in \mathbb{R}^n$*, then*

$$\mathbf{A}\underset{\sim}{x} = x_1\underset{\sim}{a}_1 + \cdots + x_n\underset{\sim}{a}_n.$$

Proof

*Proof.* Compare entry $i$ of the two sides, for each $i = 1, \ldots, m$. Entry $i$ of $\underset{\sim}{a}_j$ is $a_{ij}$, so

$$\begin{aligned} (\mathbf{A}\underset{\sim}{x})_i &= a_{i1}x_1 + \cdots + a_{in}x_n && \text{(Definition 16 in Matrices)} \\ &= x_1a_{i1} + \cdots + x_na_{in} && \text{(multiplication of numbers is commutative)} \\ &= (x_1\underset{\sim}{a}_1 + \cdots + x_n\underset{\sim}{a}_n)_i && \text{(entry } i \text{ of each } x_j\underset{\sim}{a}_j \text{ is } x_ja_{ij}) \end{aligned}$$

□

**Example 0.6** (A product as a combination of columns)**.**

$$\begin{aligned}\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}\begin{bmatrix}1\\ -1\end{bmatrix} &= 1\cdot\begin{bmatrix}1\\ 3\end{bmatrix} + (-1)\cdot\begin{bmatrix}2\\ 4\end{bmatrix} && \text{(Theorem 0.3)}\\ &= \begin{bmatrix}-1\\ -1\end{bmatrix}, && \text{(arithmetic)}\end{aligned}$$

the same answer as computing each entry as a row-times-vector dot product (Remark 4 in Matrices).

**Exercise 0.5** (Computing a matrix-vector product as a combination of columns)**.** A linear model's predictions for every observation at once are a matrix-vector product $\mathbf{A}\underset{\sim}{x}$, and Theorem 0.3 writes it as a combination of the columns of $\mathbf{A}$. Let

$$\mathbf{A} = \begin{bmatrix}2 & 1\\ 0 & 3\\ 1 & -1\end{bmatrix}, \qquad \underset{\sim}{x} = \begin{bmatrix}3\\ 2\end{bmatrix}.$$

Find $\mathbf{A}\underset{\sim}{x}$ with Theorem 0.3.

**Hint**

Use Theorem 0.3 with $x_1 = 3$ and $x_2 = 2$.

Solution

*Solution* 0.5. By Theorem 0.3,

$$\mathbf{A}\underset{\sim}{x} = 3\begin{bmatrix}2\\ 0\\ 1\end{bmatrix} + 2\begin{bmatrix}1\\ 3\\ -1\end{bmatrix} = \begin{bmatrix}6+2\\ 0+6\\ 3-2\end{bmatrix} = \begin{bmatrix}8\\ 6\\ 1\end{bmatrix}.$$

**Theorem 0.4** (A matrix maps a combination to the same combination of images)**.** *If* $\mathbf{A}$ *is an* $m \times n$ *matrix,* $\underset{\sim}{z}_1, \dots, \underset{\sim}{z}_k \in \mathbb{R}^n$ *and* $c_1, \dots, c_k \in \mathbb{R}$*, then*

$$\mathbf{A}\left(c_1\underset{\sim}{z}_1 + \cdots + c_k\underset{\sim}{z}_k\right) = c_1\,\mathbf{A}\underset{\sim}{z}_1 + \cdots + c_k\,\mathbf{A}\underset{\sim}{z}_k.$$

Proof

*Proof.* The function $\underset{\sim}{x} \mapsto \mathbf{A}\underset{\sim}{x}$ is a linear map by Theorem 12 in Matrices, and a linear map preserves linear combinations by Theorem 13 in Matrices. Applying that theorem

with $f(\underset{\sim}{x}) = \mathbf{A}\underset{\sim}{x}$ and $\underset{\sim}{v}_i = \underset{\sim}{z}_i$ gives the identity. □

**Example 0.7** (Applying a matrix to a combination)**.** Let $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $\underset{\sim}{z}_1 = (1, 0)$ and $\underset{\sim}{z}_2 = (0, 1)$. Then $\mathbf{A}\underset{\sim}{z}_1 = (1, 3)$ and $\mathbf{A}\underset{\sim}{z}_2 = (2, 4)$, and for the combination $2\underset{\sim}{z}_1 - \underset{\sim}{z}_2 = (2, -1)$:

$$\begin{aligned} \mathbf{A}\,(2, -1) &= (1 \cdot 2 + 2 \cdot (-1),\ 3 \cdot 2 + 4 \cdot (-1)) && \text{(Definition 16 in Matrices)} \\ &= (0, 2) && \text{(arithmetic)} \\ &= 2\,(1, 3) - (2, 4), && \text{(arithmetic)} \end{aligned}$$

which is $2\,\mathbf{A}\underset{\sim}{z}_1 - \mathbf{A}\underset{\sim}{z}_2$, as Theorem 0.4 says.

**Exercise 0.6** (Applying a matrix to a combination without knowing the matrix)**.** Theorem 0.4 lets a matrix-vector product of a combination be found from the matrix-vector product of each part, even when the matrix itself is not known. Suppose that a matrix $\mathbf{A}$ satisfies

$$\mathbf{A}\underset{\sim}{z}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad \mathbf{A}\underset{\sim}{z}_2 = \begin{bmatrix} 0 \\ -1 \end{bmatrix}.$$

Find $\mathbf{A}\,(3\underset{\sim}{z}_1 + 2\underset{\sim}{z}_2)$.

**Hint**

Use Theorem 0.4 with $c_1 = 3$ and $c_2 = 2$.

Solution

*Solution* 0.6. By Theorem 0.4,

$$\mathbf{A}\,(3\underset{\sim}{z}_1 + 2\underset{\sim}{z}_2) = 3\begin{bmatrix} 1 \\ 2 \end{bmatrix} + 2\begin{bmatrix} 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}.$$

**Theorem 0.5** (More than $p$ vectors in $\mathbb{R}^p$ are linearly dependent)**.** *If $k > p$, then any $k$ vectors $\underset{\sim}{v}_1, \ldots, \underset{\sim}{v}_k \in \mathbb{R}^p$ are not linearly independent (Definition 0.1): some numbers $c_1, \ldots, c_k$, not all zero, satisfy $c_1\underset{\sim}{v}_1 + \cdots + c_k\underset{\sim}{v}_k = \underset{\sim}{0}_p$.*

Proof

*Proof.* Induction on $p$.
**Base case,** $p = 1$**.** Each $\underset{\sim}{v}_i$ is a single number $v_i$, and $k \geq 2$. If $v_1 = 0$, take $c_1 = 1$ and every other $c_i = 0$. Otherwise take $c_1 = v_2$, $c_2 = -v_1$ and every other $c_i = 0$; then $c_2 \neq 0$, and $v_2 v_1 - v_1 v_2 = 0$.
**Inductive step.** Let $p \geq 2$, assume the theorem holds for vectors in $\mathbb{R}^{p-1}$, and take $k > p$ vectors in $\mathbb{R}^p$. Write $v_{i,p}$ for the last entry of $\underset{\sim}{v}_i$.
*Case 1: every* $v_{i,p} = 0$. Deleting the last entry gives $k > p-1$ vectors in $\mathbb{R}^{p-1}$, which by the inductive hypothesis have a combination with coefficients not all zero that equals $\underset{\sim}{0}_{p-1}$. The same coefficients combine the original vectors to $\underset{\sim}{0}_p$, since every last entry is 0.
*Case 2: some* $v_{i,p} \neq 0$. Renumber the vectors so that $v_{k,p} \neq 0$, and for $i = 1, \ldots, k-1$ let

$$\underset{\sim}{w}_i \stackrel{\text{def}}{=} \underset{\sim}{v}_i - \frac{v_{i,p}}{v_{k,p}} \underset{\sim}{v}_k.$$

The last entry of $\underset{\sim}{w}_i$ is $v_{i,p} - \frac{v_{i,p}}{v_{k,p}} v_{k,p} = 0$. Deleting that zero last entry gives $k-1 > p-1$ vectors in $\mathbb{R}^{p-1}$, so by the inductive hypothesis there are numbers $c_1, \ldots, c_{k-1}$, not all zero, whose combination of the shortened vectors is $\underset{\sim}{0}_{p-1}$. Putting the zero last entries back gives $c_1 \underset{\sim}{w}_1 + \cdots + c_{k-1} \underset{\sim}{w}_{k-1} = \underset{\sim}{0}_p$. Then

$$\begin{aligned}
\underset{\sim}{0}_p &= c_1 \underset{\sim}{w}_1 + \cdots + c_{k-1} \underset{\sim}{w}_{k-1} && \text{(choice of } c_1, \ldots, c_{k-1}\text{)} \\
&= \sum_{i=1}^{k-1} c_i \left( \underset{\sim}{v}_i - \frac{v_{i,p}}{v_{k,p}} \underset{\sim}{v}_k \right) && \text{(definition of } \underset{\sim}{w}_i\text{)} \\
&= \sum_{i=1}^{k-1} c_i \underset{\sim}{v}_i - \sum_{i=1}^{k-1} \frac{c_i v_{i,p}}{v_{k,p}} \underset{\sim}{v}_k && \text{(distribute each } c_i\text{, and distribute the sum over the difference} \\
&= c_1 \underset{\sim}{v}_1 + \cdots + c_{k-1} \underset{\sim}{v}_{k-1} - \left( \sum_{i=1}^{k-1} \frac{c_i v_{i,p}}{v_{k,p}} \right) \underset{\sim}{v}_k, && \text{(factor } \underset{\sim}{v}_k \text{ out of the second sum)}
\end{aligned}$$

a combination of $\underset{\sim}{v}_1, \ldots, \underset{\sim}{v}_k$ equal to $\underset{\sim}{0}_p$ whose first $k-1$ coefficients are not all zero. □

**Example 0.8** (Three vectors in $\mathbb{R}^2$)**.** Theorem 0.5 says the three vectors $(1,0)$, $(0,1)$ and $(1,1)$ in $\mathbb{R}^2$ are linearly dependent, and indeed $1 \cdot (1,0) + 1 \cdot (0,1) - 1 \cdot (1,1) = (0,0)$. The theorem says nothing about $k \leq p$ vectors: $(1,0)$ and $(0,1)$ are linearly independent, while $(1,2)$ and $(2,4)$ are not (Example 0.1).

**Exercise 0.7** (Testing four vectors in $\mathbb{R}^3$ for linear independence)**.** A table of data with more features than observations has more columns than each column has entries, and

Theorem 0.5 settles such cases without any arithmetic. Let

$$\underset{\sim}{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \underset{\sim}{v}_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}, \quad \underset{\sim}{v}_3 = \begin{bmatrix} 7 \\ 8 \\ 10 \end{bmatrix}, \quad \underset{\sim}{v}_4 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

Are these four vectors in $\mathbb{R}^3$ linearly independent?

**Hint**

Compare the number of vectors with $p = 3$ in Theorem 0.5.

Solution

*Solution* 0.7. No. There are $k = 4$ vectors in $\mathbb{R}^3$, and $4 > 3$, so by Theorem 0.5 they are linearly dependent.

**Definition 0.3** (Basis)**.** A **basis** [6] of a subspace $\mathcal{S}$ (Definition 0.1) is a list of vectors that are linearly independent (Definition 0.1) and span $\mathcal{S}$ (Definition 0.2).

**Example 0.9** (Two bases of $\mathbb{R}^2$)**.**

- $(1, 0)$ and $(0, 1)$ form a basis of $\mathbb{R}^2$: they are linearly independent, and any $(x, y) \in \mathbb{R}^2$ equals $x\,(1, 0) + y\,(0, 1)$.
- $(1, 0)$ and $(1, 1)$ also form a basis of $\mathbb{R}^2$: Example 0.1 shows they are linearly independent, and any $(x, y)$ equals $(x - y)\,(1, 0) + y\,(1, 1)$, because $(x - y) + y = x$ in the first entry and $0 + y = y$ in the second.

**Example 0.10** (Lists that are not bases of $\mathbb{R}^2$)**.**

- $(1, 0)$ and $(2, 0)$ are not a basis of $\mathbb{R}^2$. They are not linearly independent, since $2\,(1, 0) - (2, 0) = \underset{\sim}{0}$, and they do not span $\mathbb{R}^2$: every combination $c_1(1, 0) + c_2(2, 0)$ has second entry $\widetilde{0}$, so $(0, 1)$ is not in their span.
- $(1, 0)$, $(0, 1)$ and $(1, 1)$ span $\mathbb{R}^2$, but they are not a basis: $(1, 0) + (0, 1) - (1, 1) = \underset{\sim}{0}$, so they are not linearly independent.

**Exercise 0.8** (Testing whether $(1, 1)$ and $(1, -1)$ form a basis of $\mathbb{R}^2$)**.** Describing every vector in $\mathbb{R}^2$ through $(1, 1)$ and $(1, -1)$ instead of the axes works only if those two vectors form a basis. Let

$$\underset{\sim}{a}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad \underset{\sim}{a}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Is $\underset{\sim}{a}_1, \underset{\sim}{a}_2$ a basis of $\mathbb{R}^2$?

**Hint**

Check both conditions of Definition 0.3. For spanning, look for $c_1$ and $c_2$ with $c_1\underset{\sim}{a}_1 + c_2\underset{\sim}{a}_2 = (x, y)^\top$.

Solution

*Solution* 0.8. Yes. They span $\mathbb{R}^2$: for every $(x, y)^\top \in \mathbb{R}^2$,

$$\frac{x+y}{2}\begin{bmatrix} 1 \\ 1 \end{bmatrix} + \frac{x-y}{2}\begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}.$$

They are linearly independent: if $c_1\underset{\sim}{a}_1 + c_2\underset{\sim}{a}_2 = \underset{\sim}{0}$, then $c_1 + c_2 = 0$ and $c_1 - c_2 = 0$, and adding the two equations gives $c_1 = 0$ and then $c_2 = 0$. By Definition 0.3, $\underset{\sim}{a}_1, \underset{\sim}{a}_2$ is a basis of $\mathbb{R}^2$.

**Theorem 0.6** (Coefficients in a basis are unique)**.** *If $\underset{\sim}{a}_1, \ldots, \underset{\sim}{a}_k$ is a basis of a subspace $\mathcal{S}$ (Definition 0.3), then every $\underset{\sim}{x} \in \mathcal{S}$ is a linear combination $\underset{\sim}{x} = c_1\underset{\sim}{a}_1 + \cdots + c_k\underset{\sim}{a}_k$ for exactly one list of numbers $c_1, \ldots, c_k$.*

Proof

*Proof.* At least one such list exists, because the basis spans $\mathcal{S}$ (Definition 0.2). Suppose two lists both work:

$$\begin{aligned} c_1\underset{\sim}{a}_1 + \cdots + c_k\underset{\sim}{a}_k &= \underset{\sim}{x} \\ &= d_1\underset{\sim}{a}_1 + \cdots + d_k\underset{\sim}{a}_k. \end{aligned}$$

Subtracting the right side from the left side gives

$$\begin{aligned} \underset{\sim}{0} &= (c_1\underset{\sim}{a}_1 + \cdots + c_k\underset{\sim}{a}_k) - (d_1\underset{\sim}{a}_1 + \cdots + d_k\underset{\sim}{a}_k) && \text{(both combinations equal } \underset{\sim}{x}\text{)} \\ &= (c_1 - d_1)\,\underset{\sim}{a}_1 + \cdots + (c_k - d_k)\,\underset{\sim}{a}_k && \text{(regroup, and distribute each } \underset{\sim}{a}_i\text{)} \end{aligned}$$

Because $\underset{\sim}{a}_1, \ldots, \underset{\sim}{a}_k$ are linearly independent (Definition 0.1), every coefficient $c_i - d_i$ is 0, so $c_i = d_i$ for each $i$. □

**Example 0.11** (Coefficients of $(3,5)$ in two bases)**.** In the basis $(1,0),(0,1)$ of Example 0.9, $(3,5) = 3\,(1,0) + 5\,(0,1)$, and no other coefficients work. In the basis $(1,0),(1,1)$, the formula in Example 0.9 gives

$$\begin{aligned}(3,5) &= (3-5)\,(1,0) + 5\,(1,1)\\ &= -2\,(1,0) + 5\,(1,1).\end{aligned}$$

The same vector has different coefficients in different bases, but within one basis its coefficients are unique.
The list $(1,0),(0,1),(1,1)$ in Example 0.10 spans $\mathbb{R}^2$ but is not linearly independent, and the coefficients of $(3,5)$ are not unique:

$$\begin{aligned}(3,5) &= 3\,(1,0) + 5\,(0,1) + 0\,(1,1)\\ &= 0\,(1,0) + 2\,(0,1) + 3\,(1,1).\end{aligned}$$

**Exercise 0.9** (Counting the ways to write a vector in a basis)**.** The vectors $\underset{\sim}{a}_1 = (1,1)^\top$ and $\underset{\sim}{a}_2 = (1,-1)^\top$ form a basis of $\mathbb{R}^2$, and

$$\begin{bmatrix}5\\1\end{bmatrix} = 3\underset{\sim}{a}_1 + 2\underset{\sim}{a}_2.$$

How many lists of numbers $c_1, c_2$ satisfy $(5,1)^\top = c_1\underset{\sim}{a}_1 + c_2\underset{\sim}{a}_2$?

**Hint**

Use Theorem 0.6.

Solution

*Solution* 0.9. Exactly one, namely $c_1 = 3$ and $c_2 = 2$. By Theorem 0.6, a vector in the span of a basis has only one list of numbers, and $3, 2$ is a list that works.

**Definition 0.4** (Coordinates in a basis)**.** Let $\underset{\sim}{a}_1, \dots, \underset{\sim}{a}_k$ be a basis of a subspace $\mathcal{S}$ (Definition 0.3), and let $\underset{\sim}{x} \in \mathcal{S}$. The **coordinates** [7] of $\underset{\sim}{x}$ in this basis are the numbers $c_1, \dots, c_k$ with $\underset{\sim}{x} = c_1\underset{\sim}{a}_1 + \cdots + c_k\underset{\sim}{a}_k$. By Theorem 0.6 there is exactly one such list of numbers.

**Definition 0.5** (Coordinate vector)**.** With $\underset{\sim}{x}$ and its coordinates $c_1, \dots, c_k$ as in Definition 0.4, $(c_1, \dots, c_k) \in \mathbb{R}^k$ is the **coordinate vector** of $\underset{\sim}{x}$.

**Example 0.12** (Coordinates of $(4,1)$ in two bases of $\mathbb{R}^2$)**.** In the basis $(1,0),(1,1)$ of Example 0.9, the coordinates of $(4,1)$ are 3 and 1:

$$\begin{aligned} 3\,(1,0)+1\,(1,1) &= (3+1,\ 0+1) && \text{(Definition 9 in Vectors, entry by entry)} \\ &= (4,1). && \text{(add)} \end{aligned}$$

In the basis $(1,0),(0,1)$, its coordinates are its entries, 4 and 1.

**Exercise 0.10** (Finding coordinates in a basis other than the indicator vectors)**.** The coordinates of a vector depend on the basis, and they match its entries only in the basis of indicator vectors. The vectors $\underset{\sim}{a}_1 = (2,0)^\top$ and $\underset{\sim}{a}_2 = (1,1)^\top$ form a basis of $\mathbb{R}^2$. Let

$$\underset{\sim}{x} = \begin{bmatrix} 7 \\ 3 \end{bmatrix}.$$

Find the coordinates of $\underset{\sim}{x}$ in the basis $\underset{\sim}{a}_1, \underset{\sim}{a}_2$.

**Hint**

Write $c_1\underset{\sim}{a}_1 + c_2\underset{\sim}{a}_2$ entry by entry, and start from the second entry.

Solution

*Solution* 0.10. The coordinates of $\underset{\sim}{x}$ are 2 and 3. Entry by entry, $c_1\underset{\sim}{a}_1 + c_2\underset{\sim}{a}_2 = (2c_1+c_2,\ c_2)^\top$. The second entry gives $c_2 = 3$, and then the first gives $2c_1 + 3 = 7$, so $c_1 = 2$. As a check, $2(2,0)^\top + 3(1,1)^\top = (7,3)^\top$.

**Exercise 0.11** (Describing a point on a plane in $\mathbb{R}^3$ with two numbers)**.** A point on a plane through the origin in $\mathbb{R}^3$ needs only two numbers, its coordinate vector in a basis of the plane. The vectors $\underset{\sim}{a}_1 = (1,1,0)^\top$ and $\underset{\sim}{a}_2 = (0,1,1)^\top$ form a basis of a plane $\mathcal{S}$, which contains

$$\underset{\sim}{x} = \begin{bmatrix} 2 \\ 5 \\ 3 \end{bmatrix}.$$

Find the coordinate vector of $\underset{\sim}{x}$ in the basis $\underset{\sim}{a}_1, \underset{\sim}{a}_2$.

**Hint**

Write $c_1\underset{\sim}{a}_1 + c_2\underset{\sim}{a}_2$ entry by entry, then use Definition 0.5.

Solution

*Solution* 0.11. Entry by entry, $c_1\underset{\sim}{a}_1 + c_2\underset{\sim}{a}_2 = (c_1,\ c_1 + c_2,\ c_2)^\top$. Matching $\underset{\sim}{x}$ gives $c_1 = 2$ and $c_2 = 3$, and the second entry checks: $2 + 3 = 5$. By Definition 0.5, the coordinate vector of $\underset{\sim}{x}$ is $(2, 3)^\top \in \mathbb{R}^2$.

**Theorem 0.7** (All bases of a subspace have the same number of vectors)**.** *If* $\underset{\sim}{a}_1, \dots, \underset{\sim}{a}_k$ *and* $\underset{\sim}{b}_1, \dots, \underset{\sim}{b}_l$ *are both bases of a subspace* $\mathcal{S}$ *of* $\mathbb{R}^p$ *(Definition 0.3), then* $k = l$*.*

Proof

*Proof.* Let $\mathbf{A}$ be the $p \times k$ matrix with columns $\underset{\sim}{a}_1, \dots, \underset{\sim}{a}_k$, and $\mathbf{B}$ the $p \times l$ matrix with columns $\underset{\sim}{b}_1, \dots, \underset{\sim}{b}_l$.

**Write $\mathbf{A}$ in terms of $\mathbf{B}$.** Each $\underset{\sim}{a}_j$ is in $\mathcal{S}$, and the $\underset{\sim}{b}$'s span $\mathcal{S}$ (Definition 0.2), so $\underset{\sim}{a}_j = \mathbf{B}\underset{\sim}{c}_j$ for some $\underset{\sim}{c}_j \in \mathbb{R}^l$ (Definition 16 in Matrices). Let $\mathbf{C}$ be the $l \times k$ matrix with columns $\underset{\sim}{c}_1, \dots, \underset{\sim}{c}_k$. Then $\underset{p\times k}{\mathbf{A}} = \underset{p\times l}{\mathbf{B}}\ \underset{l\times k}{\mathbf{C}}$, because column $j$ of $\mathbf{BC}$ is $\mathbf{B}\underset{\sim}{c}_j$.

**The columns of $\mathbf{C}$ are linearly independent.** Suppose $\mathbf{C}\underset{\sim}{x} = \underset{\sim}{0}$ for some $\underset{\sim}{x} \in \mathbb{R}^k$. Then

$$\begin{aligned}
\mathbf{A}\underset{\sim}{x} &= (\mathbf{BC})\,\underset{\sim}{x} && \text{(substitute } \mathbf{A} = \mathbf{BC}\text{)}\\
&= \mathbf{B}\,(\mathbf{C}\underset{\sim}{x}) && \text{(Theorem 7 in Matrices)}\\
&= \mathbf{B}\,\underset{\sim}{0} && \text{(substitute } \mathbf{C}\underset{\sim}{x} = \underset{\sim}{0}\text{)}\\
&= \underset{\sim}{0} && \text{(a matrix times the zero vector is } \underset{\sim}{0}\text{)}
\end{aligned}$$

$\mathbf{A}\underset{\sim}{x}$ is the combination $x_1\underset{\sim}{a}_1 + \cdots + x_k\underset{\sim}{a}_k$ (Theorem 0.3), and the $\underset{\sim}{a}$'s are linearly independent (Definition 0.1), so $\underset{\sim}{x} = \underset{\sim}{0}$. So the only combination of the columns of $\mathbf{C}$ that equals $\underset{\sim}{0}$ has all coefficients 0.

**Count.** $\mathbf{C}$ has $k$ linearly independent columns of length $l$. More than $l$ vectors in $\mathbb{R}^l$ are never linearly independent (Theorem 0.5), so $k \le l$. Exchanging the roles of the two bases gives $l \le k$, so $k = l$. □

**Example 0.13** (Both bases of $\mathbb{R}^2$ in Example 0.9 have two vectors)**.** The bases $(1, 0), (0, 1)$ and $(1, 0), (1, 1)$ of $\mathbb{R}^2$ in Example 0.9 both have 2 vectors, as Theorem 0.7 requires. So no list of 3 vectors is a basis of $\mathbb{R}^2$, which agrees with Example 0.10: the list $(1, 0), (0, 1), (1, 1)$ spans $\mathbb{R}^2$ but is not linearly independent.

**Definition 0.6** (Dimension)**.** The **dimension** of a subspace $\mathcal{S}$, written $\dim(\mathcal{S})$, is the number of vectors in any basis of $\mathcal{S}$ (Definition 0.3). Theorem 0.7 makes this number well defined. By convention, the empty list is the basis of $\{\underset{\sim}{0}\}$, so $\dim(\{\underset{\sim}{0}\}) = 0$.

**Example 0.14** (The dimensions of some subspaces)**.**

- $\dim(\{\underset{\sim}{0}\}) = 0$, by the convention in Definition 0.6.
- $\dim(\mathbb{R}^2) = 2$, by Example 0.9. More generally, $\dim(\mathbb{R}^p) = p$: the indicator vectors $\underset{\sim}{e}_1, \ldots, \underset{\sim}{e}_p$ (Definition 18 in Vectors) are linearly independent, since $c_1\underset{\sim}{e}_1 + \cdots + c_p\underset{\sim}{e}_p = (c_1, \ldots, c_p)$, which is $\underset{\sim}{0}_p$ only if every $c_i = 0$; and they span $\mathbb{R}^p$, since any $\underset{\sim}{x} = x_1\underset{\sim}{e}_1 + \cdots + x_p\underset{\sim}{e}_p$.
- The line in Example 0.1 has dimension 1: the single vector $(1, 2)$ is linearly independent and spans the line.
- The plane $z = 0$ in Example 0.4 has dimension 2, with basis $(1, 0, 0), (0, 1, 0)$.

**Exercise 0.12** (Testing whether two vectors can form a basis of $\mathbb{R}^3$)**.** A list of two features cannot describe every point in $\mathbb{R}^3$, and Theorem 0.7 says why in terms of bases. The indicator vectors $\underset{\sim}{e}_1$, $\underset{\sim}{e}_2$ and $\underset{\sim}{e}_3$ form a basis of $\mathbb{R}^3$.
Can a list of 2 vectors be a basis of $\mathbb{R}^3$?

**Hint**

Use Theorem 0.7 with the basis $\underset{\sim}{e}_1, \underset{\sim}{e}_2, \underset{\sim}{e}_3$.

Solution

*Solution* 0.12. No. $\underset{\sim}{e}_1, \underset{\sim}{e}_2, \underset{\sim}{e}_3$ is a basis of $\mathbb{R}^3$ with 3 vectors, so by Theorem 0.7 every basis of $\mathbb{R}^3$ has 3 vectors, not 2.

**Exercise 0.13** (Finding the dimension of a span that includes a multiple)**.** The dimension of a span can be smaller than the number of vectors that span it, for example when one vector is a multiple of another. Let

$$\mathcal{S} = \operatorname{span}\left\{(1, 2, 0)^\top,\ (2, 4, 0)^\top,\ (0, 0, 1)^\top\right\}.$$

Find $\dim(\mathcal{S})$.

**Hint**

Look for a vector in the list that is a multiple of another, then check that the rest are linearly independent.

Solution

*Solution* 0.13. $\dim(\mathcal{S}) = 2$. Since $(2, 4, 0)^\top = 2(1, 2, 0)^\top$, dropping $(2, 4, 0)^\top$ leaves the span unchanged, so $(1, 2, 0)^\top$ and $(0, 0, 1)^\top$ span $\mathcal{S}$. They are linearly independent: if

$c_1(1,2,0)^\top + c_2(0,0,1)^\top = \underset{\sim}{0}$, the first entry gives $c_1 = 0$ and the third gives $c_2 = 0$. So they form a basis of $\mathcal{S}$ with 2 vectors, and by Definition 0.6 the dimension is 2.

**Theorem 0.8** (Adding a vector outside the span keeps a list independent)**.** *If $\underset{\sim}{v}_1, \dots, \underset{\sim}{v}_k$ are linearly independent (Definition 0.1) and $\underset{\sim}{w}$ is not in* $\operatorname{span}\{\underset{\sim}{v}_1, \dots, \underset{\sim}{v}_k\}$ *(Definition 0.2), then $\underset{\sim}{v}_1, \dots, \underset{\sim}{v}_k, \underset{\sim}{w}$ are linearly independent.*

Proof

*Proof.* Suppose $c_1 \underset{\sim}{v}_1 + \cdots + c_k \underset{\sim}{v}_k + d\,\underset{\sim}{w} = \underset{\sim}{0}$.
**The coefficient $d$ is 0.** If $d \neq 0$, then

$$\begin{aligned} \underset{\sim}{w} &= \frac{1}{d}\,(d\,\underset{\sim}{w}) && (d \neq 0) \\ &= \frac{1}{d}\,(-c_1 \underset{\sim}{v}_1 - \cdots - c_k \underset{\sim}{v}_k) && \text{(solve the supposed equation for } d\,\underset{\sim}{w}) \\ &= \left(-\frac{c_1}{d}\right) \underset{\sim}{v}_1 + \cdots + \left(-\frac{c_k}{d}\right) \underset{\sim}{v}_k, && \text{(distribute } \tfrac{1}{d}) \end{aligned}$$

which puts $\underset{\sim}{w}$ in $\operatorname{span}\{\underset{\sim}{v}_1, \dots, \underset{\sim}{v}_k\}$, contrary to the assumption.
**The other coefficients are 0.** With $d = 0$, the equation reads $c_1 \underset{\sim}{v}_1 + \cdots + c_k \underset{\sim}{v}_k = \underset{\sim}{0}$, so every $c_i = 0$, because $\underset{\sim}{v}_1, \dots, \underset{\sim}{v}_k$ are linearly independent. □

**Example 0.15** (Adding $(0,0,1)$ to two independent vectors in $\mathbb{R}^3$)**.** The vectors $(1,0,0)$ and $(0,1,0)$ are linearly independent, and their span is the plane $z = 0$ (Example 0.4). The vector $(0,0,1)$ has third entry 1, so it is not in that plane, and by Theorem 0.8 the three vectors $(1,0,0)$, $(0,1,0)$, $(0,0,1)$ are linearly independent. Adding $(1,1,0)$ instead would not work: $(1,1,0)$ is in the span, and $(1,0,0) + (0,1,0) - (1,1,0) = \underset{\sim}{0}$.

**Exercise 0.14** (Adding a third feature to an independent pair)**.** Adding a new feature keeps a list of features linearly independent when the new one is not a combination of the old ones, which is what Theorem 0.8 says. Let

$$\underset{\sim}{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \qquad \underset{\sim}{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \qquad \underset{\sim}{w} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

Are $\underset{\sim}{v}_1$, $\underset{\sim}{v}_2$ and $\underset{\sim}{w}$ linearly independent?

**Hint**

Check whether $\underset{\sim}{w}$ is in $\operatorname{span}\{\underset{\sim}{v}_1, \underset{\sim}{v}_2\}$, then use Theorem 0.8.

Solution

*Solution* 0.14. Yes. $\underset{\sim}{v}_1$ and $\underset{\sim}{v}_2$ are linearly independent: $c_1\underset{\sim}{v}_1 + c_2\underset{\sim}{v}_2 = (c_1, c_2, 0)^\top$ is $\underset{\sim}{0}$ only for $c_1 = c_2 = 0$. Every vector in their span has third entry 0, and $\underset{\sim}{w}$ has third entry 1, so $\underset{\sim}{w}$ is not in the span. By Theorem 0.8, $\underset{\sim}{v}_1$, $\underset{\sim}{v}_2$ and $\underset{\sim}{w}$ are linearly independent.

**Theorem 0.9** (An independent list extends to a basis)**.** *If $\mathcal{S}$ is a subspace of $\mathbb{R}^p$ (Definition 0.1) and $\underset{\sim}{v}_1, \ldots, \underset{\sim}{v}_k \in \mathcal{S}$ are linearly independent (Definition 0.1), then adding vectors of $\mathcal{S}$ to the list gives a basis of $\mathcal{S}$ (Definition 0.3). Starting from the empty list shows that every subspace has a basis.*

Proof

*Proof.* Repeat the following step. If the current list spans $\mathcal{S}$, stop: it is linearly independent and spans $\mathcal{S}$, so it is a basis. Otherwise, some $\underset{\sim}{w} \in \mathcal{S}$ is not in the span of the current list; add $\underset{\sim}{w}$ to the list, which stays linearly independent by Theorem 0.8.
Every list produced is a linearly independent list of vectors in $\mathbb{R}^p$, so it has at most $p$ vectors (Theorem 0.5). Each step adds one vector, so the process stops after at most $p - k$ steps, and when it stops the list is a basis of $\mathcal{S}$.
For the empty list, which is linearly independent and spans $\{\underset{\sim}{0}\}$ by the conventions in Definition 0.2, the first step asks whether $\mathcal{S} = \{\underset{\sim}{0}\}$: if so, the empty list is already a basis; if not, the process adds vectors as described. □

**Example 0.16** (Extending $(1, 1, 0)$ to a basis of $\mathbb{R}^3$)**.** Start with the single vector $(1, 1, 0)$.

1. $(1, 0, 0)$ is not in $\text{span}\{(1, 1, 0)\}$, because every multiple $c\,(1, 1, 0)$ has equal first and second entries. Add it.
2. $(0, 0, 1)$ is not in $\text{span}\{(1, 1, 0), (1, 0, 0)\}$, because every combination of those two vectors has third entry 0. Add it.
3. The list $(1, 1, 0), (1, 0, 0), (0, 0, 1)$ spans $\mathbb{R}^3$: $(x, y, z) = y\,(1, 1, 0) + (x - y)\,(1, 0, 0) + z\,(0, 0, 1)$. Stop.

The result is a basis of $\mathbb{R}^3$ that contains the starting vector.

**Exercise 0.15** (Extending one vector to a basis of $\mathbb{R}^3$)**.** Theorem 0.9 says that any linearly independent list can be completed to a basis. Let

$$\underset{\sim}{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}.$$

Find two indicator vectors that extend $\underset{\sim}{v}_1$ to a basis of $\mathbb{R}^3$.

**Hint**

Add indicator vectors one at a time, each outside the span of the list so far, as Theorem 0.8 allows.

Solution

*Solution* 0.15. One answer is $\underset{\sim}{e}_2$ and $\underset{\sim}{e}_3$. The list $\underset{\sim}{v}_1$ is linearly independent, since $\underset{\sim}{v}_1 \neq \underset{\sim}{0}$. $\underset{\sim}{e}_2 = (0,1,0)^\top$ is not a multiple of $\underset{\sim}{v}_1$, whose first two entries match, so by Theorem 0.8 the list $\underset{\sim}{v}_1, \underset{\sim}{e}_2$ is linearly independent. Every vector in its span has third entry 0, so $\underset{\sim}{e}_3$ is outside it, and $\underset{\sim}{v}_1, \underset{\sim}{e}_2, \underset{\sim}{e}_3$ is linearly independent too. These three vectors span $\mathbb{R}^3$, because $(x,y,z)^\top = x\underset{\sim}{v}_1 + (y-x)\underset{\sim}{e}_2 + z\underset{\sim}{e}_3$, so they form a basis of $\mathbb{R}^3$. The pair $\underset{\sim}{e}_1$ and $\underset{\sim}{e}_3$ works too.

**Theorem 0.10** (An independent list in a subspace has at most dim vectors)**.** *Let $\mathcal{S}$ be a subspace of $\mathbb{R}^p$ with $\dim(\mathcal{S}) = d$ (Definition 0.6).*

1. *Any linearly independent list of vectors in $\mathcal{S}$ has at most $d$ vectors.*
2. *If $\mathcal{T}$ is a subspace with $\mathcal{T} \subseteq \mathcal{S}$, then $\dim(\mathcal{T}) \leq \dim(\mathcal{S})$.*

Proof

*Proof.* **Part 1.** If $d = 0$, then $\mathcal{S} = \{\underset{\sim}{0}\}$, so every vector in a nonempty list from $\mathcal{S}$ is $\underset{\sim}{0}$, and $1 \cdot \underset{\sim}{0} = \underset{\sim}{0}$ shows the list is not linearly independent; only the empty list, with $0 = d$ vectors, is. Now let $d \geq 1$, and let $\underset{\sim}{a}_1, \ldots, \underset{\sim}{a}_d$ be a basis of $\mathcal{S}$, and let $\underset{\sim}{u}_1, \ldots, \underset{\sim}{u}_k \in \mathcal{S}$ be linearly independent. Each $\underset{\sim}{u}_j$ is in $\mathcal{S}$, so $\underset{\sim}{u}_j = c_{1j}\underset{\sim}{a}_1 + \cdots + c_{dj}\underset{\sim}{a}_d$ for some coordinate vector $\underset{\sim}{c}_j = (c_{1j}, \ldots, c_{dj}) \in \mathbb{R}^d$. Suppose $k > d$. Then $\underset{\sim}{c}_1, \ldots, \underset{\sim}{c}_k$ are $k > d$ vectors in $\mathbb{R}^d$, so some $x_1, \ldots, x_k$, not all zero, satisfy $x_1\underset{\sim}{c}_1 + \cdots + x_k\underset{\sim}{c}_k = \underset{\sim}{0}_d$ (Theorem 0.5); entry $i$ of that equation says $\sum_{j=1}^{k} x_j c_{ij} = 0$. Then

$$
\begin{aligned}
\sum_{j=1}^{k} x_j \underset{\sim}{u}_j &= \sum_{j=1}^{k} x_j \sum_{i=1}^{d} c_{ij} \underset{\sim}{a}_i && \text{(substitute each } \underset{\sim}{u}_j\text{)} \\
&= \sum_{j=1}^{k} \sum_{i=1}^{d} x_j c_{ij}\, \underset{\sim}{a}_i && \text{(distribute each } x_j \text{ over the inner sum)} \\
&= \sum_{i=1}^{d} \sum_{j=1}^{k} x_j c_{ij}\, \underset{\sim}{a}_i && \text{(swap the order of the finite sums)} \\
&= \sum_{i=1}^{d} \left( \sum_{j=1}^{k} x_j c_{ij} \right) \underset{\sim}{a}_i && \text{(factor } \underset{\sim}{a}_i \text{ out of the inner sum)} \\
&= \sum_{i=1}^{d} 0 \cdot \underset{\sim}{a}_i && \text{(entry } i \text{ of } \textstyle\sum_j x_j \underset{\sim}{c}_j = \underset{\sim}{0}_d\text{)} \\
&= \underset{\sim}{0}_p, && \text{(a zero multiple of a vector is } \underset{\sim}{0}\text{)}
\end{aligned}
$$

a combination of $\underset{\sim}{u}_1, \ldots, \underset{\sim}{u}_k$ with coefficients not all zero that equals $\underset{\sim}{0}_p$. That contradicts their linear independence, so $k \leq d$.

**Part 2.** $\mathcal{T}$ has a basis (Theorem 0.9) with $\dim(\mathcal{T})$ vectors. Those vectors are linearly independent and lie in $\mathcal{S}$, so by Part 1 there are at most $\dim(\mathcal{S})$ of them. □

**Example 0.17** (Subspaces of a plane have dimension at most two)**.** The plane $\mathcal{S} = \{(x, y, 0) : x, y \in \mathbb{R}\}$ in $\mathbb{R}^3$ has dimension 2 (Example 0.14). By Theorem 0.10, no three vectors with third entry 0 are linearly independent; for instance, $(1, 0, 0) + (0, 1, 0) - (1, 1, 0) = \underset{\sim}{0}$. The line $\mathcal{T} = \text{span}\{(1, 1, 0)\}$ lies in $\mathcal{S}$, and

$$
\begin{aligned}
\dim(\mathcal{T}) &= 1 \\
&\leq 2 \\
&= \dim(\mathcal{S}).
\end{aligned}
$$

**Exercise 0.16** (Testing whether four independent vectors fit in a three-dimensional subspace)**.** Theorem 0.10 caps how many linearly independent vectors a subspace can hold. Let $\mathcal{S}$ be a subspace of $\mathbb{R}^5$ with dimension 3.
Can $\mathcal{S}$ contain 4 linearly independent vectors?

**Hint**

Use part 1 of Theorem 0.10 with $d = 3$.

Solution

*Solution* 0.16. No. By part 1 of Theorem 0.10, a linearly independent list of vectors in $\mathcal{S}$ has at most $\dim(\mathcal{S}) = 3$ vectors, so it cannot have 4.

**Theorem 0.11** (A subspace inside another of the same dimension equals it)**.** *If $\mathcal{T}$ and $\mathcal{S}$ are subspaces of $\mathbb{R}^p$ with $\mathcal{T} \subseteq \mathcal{S}$ and $\dim(\mathcal{T}) = \dim(\mathcal{S})$ (Definition 0.6), then $\mathcal{T} = \mathcal{S}$.*

Proof

*Proof.* Let $d = \dim(\mathcal{S})$, and let $\underset{\sim}{t}_1, \dots, \underset{\sim}{t}_d$ be a basis of $\mathcal{T}$ (Theorem 0.9); it has $d$ vectors because $\dim(\mathcal{T}) = d$. (When $d = 0$ it is the empty list, whose span is $\{\underset{\sim}{0}\}$ by the convention in Definition 0.2, and the argument is unchanged.) Suppose some $\underset{\sim}{s} \in \mathcal{S}$ is not in $\operatorname{span}\{\underset{\sim}{t}_1, \dots, \underset{\sim}{t}_d\} = \mathcal{T}$. Then $\underset{\sim}{t}_1, \dots, \underset{\sim}{t}_d, \underset{\sim}{s}$ are linearly independent (Theorem 0.8), and they are $d + 1$ vectors of $\mathcal{S}$, which contradicts part 1 of Theorem 0.10. So every vector of $\mathcal{S}$ is in $\mathcal{T}$, that is, $\mathcal{S} \subseteq \mathcal{T}$. Together with $\mathcal{T} \subseteq \mathcal{S}$, this inclusion gives $\mathcal{T} = \mathcal{S}$. □

**Example 0.18** (Two independent vectors in a plane span it)**.** Let $\mathcal{S} = \{(x, y, 0) : x, y \in \mathbb{R}\}$, the plane $z = 0$, which has dimension 2 (Example 0.14), and let $\mathcal{T} = \operatorname{span}\{(1, 0, 0), (1, 1, 0)\}$. Both vectors have third entry 0, so $\mathcal{T} \subseteq \mathcal{S}$. They are linearly independent: $c_1(1, 0, 0) + c_2(1, 1, 0) = (c_1 + c_2, c_2, 0)$, which is $\underset{\sim}{0}$ only if $c_2 = 0$ and then $c_1 = 0$. So $\dim(\mathcal{T}) = 2$, and Theorem 0.11 gives $\mathcal{T} = \mathcal{S}$ without solving for any coefficients. As a check, $(x, y, 0) = (x - y)\,(1, 0, 0) + y\,(1, 1, 0)$.

**Example 0.19** (Equal dimensions without containment)**.** The lines $\operatorname{span}\{(1, 0, 0)\}$ and $\operatorname{span}\{(0, 1, 0)\}$ in $\mathbb{R}^3$ both have dimension 1, but they are not equal: $(1, 0, 0)$ is in the first and not the second. Theorem 0.11 does not apply, because neither line contains the other.

**Exercise 0.17** (Showing that a span of two vectors is all of $\mathbb{R}^2$)**.** Theorem 0.11 shows that two subspaces coincide by comparing dimensions instead of checking every vector. Let

$$\mathcal{T} = \operatorname{span}\Big\{(1, 2)^\top,\ (3, 1)^\top\Big\}.$$

Is $\mathcal{T}$ all of $\mathbb{R}^2$?

**Hint**

Find $\dim(\mathcal{T})$ and $\dim(\mathbb{R}^2)$, then use Theorem 0.11 with $\mathcal{S} = \mathbb{R}^2$.

Solution

*Solution* 0.17. Yes. The two vectors are linearly independent: if $c_1(1,2)^\top + c_2(3,1)^\top = \underset{\sim}{0}$, then $c_1 + 3c_2 = 0$ and $2c_1 + c_2 = 0$. Subtracting twice the first equation from the second gives $-5c_2 = 0$, so $c_2 = 0$ and then $c_1 = 0$. So they form a basis of $\mathcal{T}$, and $\dim(\mathcal{T}) = 2$. The indicator vectors $\underset{\sim}{e}_1, \underset{\sim}{e}_2$ form a basis of $\mathbb{R}^2$, so $\dim(\mathbb{R}^2) = 2$. Since $\mathcal{T} \subseteq \mathbb{R}^2$ and the dimensions agree, Theorem 0.11 gives $\mathcal{T} = \mathbb{R}^2$.


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